

WASP-50b

R-0895

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_250924 · created 2026-07-29T02:45:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460942.95 +/- 0.1 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.052
Transit depth (Rp/Rs)^2	3.07 +/- 1.81 [%]
Orbital Inclination (inc)	88.68 +/- 2.86
Airmass coefficient 1 (a1)	0.0164 +/- 0.0085
Airmass coefficient 2 (a2)	2.7 +/- 0.29
Scatter in the residuals of the lightcurve fit is	38.03 %
Best Comparison Star	#2 - [42, 39]
Optimal Aperture	5.74
Optimal Annulus	12.91
Transit Duration (day)	0.044 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.175 $\pm$ 0.052 vs 0.1404 (deviation 0.15 σ)
Duration fit vs published	0.044 d vs 0.0754 d (deviation 0.19 σ)
Mid-time O–C	16.5 min
Depth SNR	0.8
Residual scatter	38.03 %

Light curve

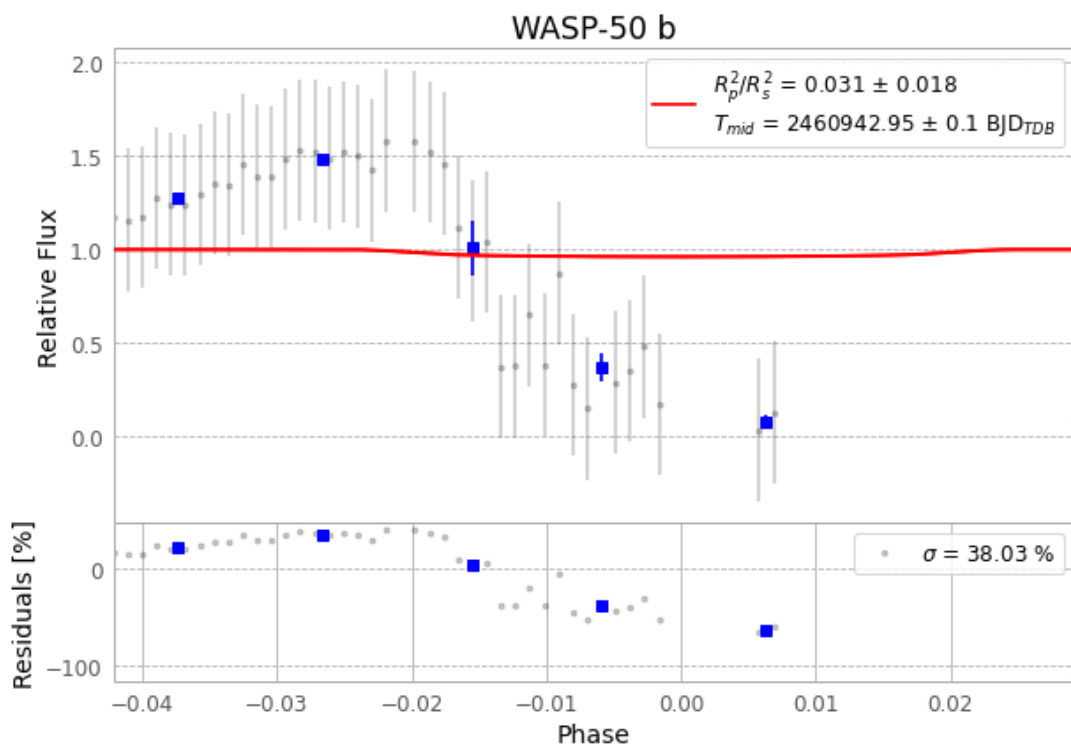


Figure 1 — FinalLightCurve_WASP-50 b_2025-09-24.png (SHA-256 80200e81b8e60835...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [234, 479], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20185815cab03b01...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

75 FITS frames (49 MB), WASP-50250924083916.FITS → WASP-50250924122115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-250924.log (1ae34bb586b2...), FinalLightCurve_WASP-50 b_2025-09-24.png (80200e81b8e6...), FinalLightCurve_WASP-50 b_2025-09-24.csv (38f0d2127e70...)

Compute resources

Wall time 576.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 02:45Z.